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CHM 414 — Past Questions & Materials
by Renance

Course: CHM 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200, 400

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Paper 1 of 5 — CHM 414 - Heterogenous Catalysis

Level 400 • First Semester • PastQuestion

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CHM 414: HETEROGENEOUS CATALYSIS (2 Credits)

Instructions: Answer Question 1 and TWO others

Time: 1 hour 45 minutes

Question 1 (30 marks)

- Write out the mathematical equations for the Freundlich and Langmuir adsorption isotherms clearly defining each parameter (4 marks).
- The volume of nitrogen gas adsorbed at STP to form a monolayer on a porous solid surface is $22.4 \text{ cm}^3/\text{g}$. If the area occupied by the nitrogen gas molecule is $16.2 \text{ \AA}$. Calculate the surface area in cm^2/g (4 marks).
- In heterogeneous catalysis, there are four essential stages. List each step (4 marks).
- List three essential properties of a good catalyst (3 marks).
- Name two groups into which the phenomenon of catalysis can be divided giving an example of each group with chemical equations only (2 marks).
- Many gas reactions are catalysed by solids, and the process starts with the gas molecules being adsorbed on to the metal surface. Explain the difference between the words *adsorbed* and *absorbed*. (2 marks).
- The adsorption of a gas on a solid surface was found to follow a Langmuir isotherm with $K = 4 \text{ Pa}^{-1}$ at a temperature of 25°C . What pressure will be required to achieve a fractional coverage of 90%? Give your answer in Pa to two decimal places (4 marks).
- In the adsorption of sugar from an aqueous solution to activated alumina, identify the adsorbent and the adsorbate. (3 marks).
- In chemical adsorption (chemisorption), what forces hold the gas molecules on the solid surface? (2 marks).
- What is the role of desorption in the process of catalysis? (2 marks).

Question 2 (20 marks)

- The adsorption of a gas on a solid surface follows the Langmuir isotherm. At a pressure of 200 Pa, 50% of the surface is covered with gas. What will be the surface coverage at a pressure of 300 Pa? (5 marks).
- Irvin Langmuir measured the following data for the adsorption of methane onto 1 gram of mica at $T = 293\text{K}$:

P (bars)	13.4	11.1	9.6	8.55	7.4	6.68	5.85
V (mm^3)	85.0	80.4	75.9	71.6	67.9	64.2	61.2

Where P is the pressure of the methane above the mica surface at equilibrium, and V is the volume of gas adsorbed onto the surface, measured at $T = 293\text{K}$ and at a pressure of 1 atm. From an appropriate graph of the data, determine V_∞ and K. (15 marks).

Question 3 (20 marks)

- A plot of $\log x/m$ against $\log P$ (where P is pressure of gas in atm) is a straight line with slope equal to 4.5 and intercept equal to 0.602. Calculate the extent of adsorption of the gas at a pressure of 1 atm. (5 marks).
- Check the applicability of Freundlich adsorption isotherm for the adsorption of solute on a solid at 25°C and find the values of K and n. (15 marks).

P (atm)	40	165	350	545	760	910	970
x/m (g)	45.5	91.5	113	121	125	126	126

Question 4 (20 marks)

- The applications of organometallic compounds in homogeneous catalysis transcend the boundaries of industry to meet the day-to-day synthesis in laboratory scale reactions. The alkene isomerization and alkene hydrogenation are two of such applications of homogeneous catalysis by the transition metal organometallic complexes. Give detailed mechanisms of both reactions indicating the different pathways and catalysts used in each case. (10 marks).
- Derive Langmuir adsorption isotherm for the following cases:
 - Molecular adsorption of CO
 - Dissociative adsorption of CO
 - Competitive adsorption of molecularly adsorbed CO and dissociatively adsorbed H_2 , without further reaction.

Suppose the catalytic formation of methanol from CO and H_2 occurs through a mechanism in which the reaction between adsorbed CO and the first H-atom determines the rate (the reverse reaction may be ignored), while all subsequent reaction steps are fast, except for the desorption of methanol, which may be considered at equilibrium with the gas phase. Devise the mechanism and derive an expression for the rate of reaction. (10 marks).

Paper 2 of 5 — CHM 414 - 2024/2025 - First Semester - Heterogeneous Catalysis

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2024/2025 ACADEMIC SESSION
CHM 414: HETEROGENEOUS CATALYSIS (2 Credits)

Time: 1 hour 45 minutes

Instructions: Answer Question 1 and TWO others

Question 1 (30 marks)

- Write out the mathematical equations for the Freundlich and Langmuir adsorption isotherms clearly defining each parameter (4 marks).
- The volume of nitrogen gas adsorbed at STP to form a monolayer on a porous solid surface is 22.4 cm³/g. If the area occupied by the nitrogen gas molecule is 16.2 Å. Calculate the surface area in cm²/g. (4 marks).
- In heterogeneous catalysis, there are four essential stages. List each step (4 marks).
- List three essential properties of a good catalyst (3 marks).
- Name two groups into which the phenomenon of catalysis can be divided giving an example of each group with chemical equations only (2 marks).
- Many gas reactions are catalysed by solids, and the process starts with the gas molecules being adsorbed on to the metal surface. Explain the difference between the words *adsorbed* and *absorbed*. (2 marks).
- The adsorption of a gas on a solid surface was found to follow a Langmuir isotherm with $K = 4 \text{ Pa}^{-1}$ at a temperature of 25 °C. What pressure will be required to achieve a fractional coverage of 90%? Give your answer in Pa to two decimal places (4 marks).
- In the adsorption of sugar from an aqueous solution to activated alumina, identify the adsorbent and the adsorbate. (3 marks).
- In chemical adsorption (chemisorption), what forces hold the gas molecules on the solid surface? (2 marks).
- What is the role of desorption in the process of catalysis? (2 marks).

Question 2 (20 marks)

- The adsorption of a gas on a solid surface follows the Langmuir isotherm. At a pressure of 200 Pa, 50% of the surface is covered with gas. What will be the surface coverage at a pressure of 300 Pa? (5 marks).
- Irvin Langmuir measured the following data for the adsorption of methane onto 1 gram of mica at $T = 90\text{K}$:

P (bars)	13.4	11.1	9.6	8.55	7.4	6.68	5.85
V (mm ³)	85.0	80.4	75.9	71.6	67.9	64.2	61.2

Where P is the pressure of the methane above the mica surface at equilibrium, and V is the volume of gas adsorbed onto the surface, measured at $T = 293\text{K}$ and at a pressure of 1 atm. From an appropriate graph of the data, determine V_m and K. (15 marks).

Question 3 (20 marks)

- A plot of $\log x/m$ against $\log P$ (where P is pressure of gas in atm) is a straight line with slope equal to 45 and intercept equal to 0.602. Calculate the extent of adsorption of the gas at a pressure of 1 atm. (5 marks).
- Check the applicability of Freundlich adsorption isotherm for the adsorption of solute on a solid at 25 °C and find the values of K and n. (15 marks).

P (atm)	40	165	350	545	760	910	970
x/m (g)	45.5	91.5	113	121	125	126	126

Question 4 (20 marks)

- The applications of organometallic compounds in homogeneous catalysis transcend the boundaries of industry to meet the day-to-day synthesis in laboratory scale reactions. The alkene isomerization and alkene hydrogenation are two of such applications of homogeneous catalysis by the transition metal organometallic complexes. Give detailed mechanisms of both reactions indicating the different pathways and catalysts used in each case. (10 marks).
- Derive Langmuir' adsorption isotherm for the following cases:
 - Molecular adsorption of CO
 - Dissociative adsorption of CO
 - Competitive adsorption of molecularly adsorbed CO and dissociatively adsorbed H₂, without further reaction.
 - Suppose the catalytic formation of methanol from CO and H₂ occurs through a mechanism in which the reaction between adsorbed CO and the first H-atom determines the rate (the reverse reaction may be ignored), while all subsequent reaction steps are fast, except for the desorption of methanol, which may be considered at equilibrium with the gas phase. Devise the mechanism and derive an expression for the rate of reaction. (10 marks).

Paper 3 of 5 — Chm 414 heterogeneous catalysis 2024/2025

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2024/2025 ACADEMIC SESSION
CHM 414: HETEROGENEOUS CATALYSIS (2 Credits)

Instructions: Answer Question 1 and TWO others Time: 1 hour 45 minutes

Question 1 (30 marks)

a) Write out the mathematical equations for the Freundlich and Langmuir adsorption isotherms clearly defining each parameter (4 marks).

b) The volume of nitrogen gas adsorbed at STP to form a monolayer on a porous solid surface is 22.4 cm³/g. If the area occupied by the nitrogen gas molecule is 16.2 Å. Calculate the surface area in cm²/g. (4 marks).

c) In heterogeneous catalysis, there are four essential stages. List each step (4 marks).

d) List three essential properties of a good catalyst (3 marks).

e) Name two groups into which the phenomenon of catalysis can be divided giving an example of each group with chemical equations only (2 marks).

f) Many gas reactions are catalysed by solids, and the process starts with the gas molecules being adsorbed on to the metal surface. Explain the difference between the words *adsorbed* and *absorbed*. (2 marks).

g) The adsorption of a gas on a solid surface was found to follow a Langmuir isotherm with $K = 4 \text{ Pa}^{-1}$ at a temperature of 25 °C. What pressure will be required to achieve a fractional coverage of 90%? Give your answer in Pa to two decimal places (4 marks).

h) In the adsorption of sugar from an aqueous solution to activated alumina, identify the adsorbent and the adsorbate. (3 marks).

i) In chemical adsorption (chemisorption), what forces hold the gas molecules on the solid surface? (2 marks).

j) What is the role of desorption in the process of catalysis? (2 marks).

Question 2 (20 marks)

a) The adsorption of a gas on a solid surface follows the Langmuir isotherm. At a pressure of 200 Pa, 50% of the surface is covered with gas. What will be the surface coverage at a pressure of 300 Pa? (5 marks).

b) Irvin Langmuir measured the following data for the adsorption of methane onto 1 gram of mica at $T = 90\text{K}$:

P (bars)	13.4	11.1	9.6	8.55	7.4	6.68	5.85
V (mm ³)	85.0	80.4	75.9	71.6	67.9	64.2	61.2

Where P is the pressure of the methane above the mica surface at equilibrium, and V is the volume of gas adsorbed onto the surface, measured at $T=293\text{K}$ and at a pressure of 1 atm. From an appropriate graph of the data, determine V_m and K. (15 marks).

Question 3 (20 marks)

a) A plot of $\log x/m$ against $\log P$ (where P is pressure of gas in atm) is a straight line with slope equal to 45 and intercept equal to 0.602. Calculate the extent of adsorption of the gas at a pressure of 1 atm. (5 marks).

b) Check the applicability of Freundlich adsorption isotherm for the adsorption of solute on a solid at 25 °C and find the values of K and n. (15 marks).

P (atm)	40	165	350	545	760	910	970
x/m (g)	45.5	91.5	113	121	125	126	126

Question 4 (20 marks)

a) The applications of organometallic compounds in homogeneous catalysis transcend the boundaries of industry to meet the day-to-day synthesis in laboratory scale reactions. The alkene isomerization and alkene hydrogenation are two of such applications of homogeneous catalysis by the transition metal organometallic complexes. Give detailed mechanisms of both reactions indicating the different pathways and catalysts used in each case. (10 marks).

b) Derive Langmuir adsorption isotherm for the following cases:

i) Molecular adsorption of CO (ii) Dissociative adsorption of CO (iii) Competitive adsorption of molecularly adsorbed CO and dissociatively adsorbed H₂, without further reaction. (iv) Suppose the catalytic formation of methanol from CO and H₂ occurs through a mechanism in which the reaction between adsorbed CO and the first H-atom determines the rate (the reverse reaction may be ignored), while all subsequent reaction steps are fast, except for the desorption of methanol, which may be considered at equilibrium with the gas phase. Devise the mechanism and derive an expression for the rate of reaction. (10 marks).

Paper 4 of 5 — CHM 414 HETEROGENEOUS CATALYSIS 2024 2025

Level 400 • First Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2024/2025 ACADEMIC SESSION
CHM 414: HETEROGENEOUS CATALYSIS (2 Credits)

Instructions: Answer Question 1 and TWO others

Time: 1 hour 45 minutes

Question 1 (30 marks)

- Write out the mathematical equations for the Freundlich and Langmuir adsorption isotherms clearly defining each parameter (4 marks).
- The volume of nitrogen gas adsorbed at STP to form a monolayer on a porous solid surface is 22.4 cm³/g. If the area occupied by the nitrogen gas molecule is 16.2 Å. Calculate the surface area in cm²/g. (4 marks).
- In heterogeneous catalysis, there are four essential stages. List each step (4 marks).
- List three essential properties of a good catalyst (3 marks).
- Name two groups into which the phenomenon of catalysis can be divided giving an example of each group with chemical equations only (2 marks).
- Many gas reactions are catalysed by solids, and the process starts with the gas molecules being adsorbed on to the metal surface. Explain the difference between the words *adsorbed* and *absorbed*. (2 marks).
- The adsorption of a gas on a solid surface was found to follow a Langmuir isotherm with $K = 4 \text{ Pa}^{-1}$ at a temperature of 25 °C. What pressure will be required to achieve a fractional coverage of 90%? Give your answer in Pa to two decimal places (4 marks).
- In the adsorption of sugar from an aqueous solution to activated alumina, identify the adsorbent and the adsorbate. (3 marks).
- In chemical adsorption (chemisorption), what forces hold the gas molecules on the solid surface? (2 marks).
- What is the role of desorption in the process of catalysis? (2 marks).

Question 2 (20 marks)

- The adsorption of a gas on a solid surface follows the Langmuir isotherm. At a pressure of 200 Pa, 50% of the surface is covered with gas. What will be the surface coverage at a pressure of 300 Pa? (5 marks).
- Irvin Langmuir measured the following data for the adsorption of methane onto 1 gram of mica at $T = 293 \text{ K}$:

P (bars)	13.4	11.1	9.6	8.55	7.4	6.68	5.85
V (mm ³)	85.0	80.4	75.9	71.6	67.9	64.2	61.2

Where P is the pressure of the methane above the mica surface at equilibrium, and V is the volume of gas adsorbed onto the surface, measured at $T = 293 \text{ K}$ and at a pressure of 1 atm. From an appropriate graph of the data, determine V_m and K. (15 marks).

Question 3 (20 marks)

- A plot of $\log x/m$ against $\log P$ (where P is pressure of gas in atm) is a straight line with slope equal to 45 and intercept equal to 0.602. Calculate the extent of adsorption of the gas at a pressure of 1 atm. (5 marks).
- Check the applicability of Freundlich adsorption isotherm for the adsorption of solute on a solid at 25 °C and find the values of K and n. (15 marks).

P (atm)	40	165	350	545	760	910	970
x/m (g)	45.5	91.5	113	121	125	126	126

Question 4 (20 marks)

- The applications of organometallic compounds in homogeneous catalysis transcend the boundaries of industry to meet the day-to-day synthesis in laboratory scale reactions. The alkene isomerization and alkene hydrogenation are two of such applications of homogeneous catalysis by the transition metal organometallic complexes. Give detailed mechanisms of both reactions indicating the different pathways and catalysts used in each case. (10 marks).
- Derive Langmuir adsorption isotherm for the following cases:
 - Molecular adsorption of CO
 - Dissociative adsorption of CO
 - Competitive adsorption of molecularly adsorbed CO and dissociatively adsorbed H₂, without further reaction.
 - Suppose the catalytic formation of methanol from CO and H₂ occurs through a mechanism in which the reaction between adsorbed CO and the first H-atom determines the rate (the reverse reaction may be ignored), while all subsequent reaction steps are fast, except for the desorption of methanol, which may be considered at equilibrium with the gas phase. Devise the mechanism and derive an expression for the rate of reaction. (10 marks).

END

Paper 5 of 5 — CHM 414

Level 400 • First Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2019/2020 ACADEMIC SESSION
CHM 414: HETEROGENEOUS CATALYSIS (2 Credits)

Instructions: Answer Question 1 and TWO others **Time:** 2 hours

Question 1

- Identify the type of catalysis (homogeneous or heterogeneous) and specify the catalysts in the following reactions: (i) Haber process (ii) Contact process (iii) Hydrogenation of alkenes
- Adsorption of methane follows the Langmuir adsorption isotherm at 90 K. If $p = 1.896 \text{ cm}^3/\text{g}^{-1} \text{ bar}^{-1}$ and $b = 0.146 \text{ bar}^{-1}$. Calculate the value of θ .
- Explain the difference between the activity and selectivity of a catalyst.
- Calculate the adsorption of a dye on activated carbon at 25 °C where $k = 0.025$, $n = 0.5$ and $c = 0.04$.
- The adsorption of sulphur dioxide by polymer pellets at 0 °C is shown in the table below.

pSO ₂ (mmHg)	5	10	15	20	30	40	50	60	70
Uptake (mg mol/g)	1.75	2.20	2.40	2.62	2.75	2.85	3.00	3.05	3.12

Using these data, determine the:

- Langmuir constants (ii) Freundlich constants (iii) Do these two isotherms fit the data well?

Question 2

- In a solution of 100 ml, 0.5M acetic acid, 1 g of active charcoal is added which adsorbs acetic acid. It is found that the concentration of acetic acid becomes 0.49 M. If the surface area of charcoal is $3.01 \times 10^2 \text{ m}^2$, calculate the area occupied by a single acetic acid molecule on the surface of the charcoal.
- Distinguish between homogeneous catalysis and heterogeneous catalysis. Describe three important industrial processes that utilize heterogeneous catalysis.
- State the role of desorption in the process of catalysis.

Question 3

- Which adsorption isotherm is applicable to physical adsorption? Give reasons for your answer.
- Check the applicability of Freundlich adsorption isotherm for the adsorption of solute on a solid at 25 °C and find the values of K and n.

Concentration (mol/L)	0.05	0.10	0.50	1.00	1.50
x/m (g)	0.048	0.072	0.144	0.192	0.228

- Different concentrations of aqueous solutions placed with charcoal which adsorbs a part of solute from solution is reported below at equilibrium

Concentration $\times 10^{-2}$	2.0	4.0
x/m (g)	0.185	0.290

Calculate the values of Log K and n.

Question 4

- Derive the Langmuir adsorption theorem for the reaction $A(g) + B(s) \rightleftharpoons AB$. Show all the various steps taken.
- The following data were obtained for the extent of adsorption of acetone on charcoal.

Acetone at equilibrium (mmol/L)	15.0	42.0	69.0	86.1	106.2
mmol acetone adsorbed per g charcoal (mmol/g)	0.60	1.05	1.55	2.05	2.60

- Calculate the Langmuir parameters
- What does the parameter, K, mean physically?

- For an adsorbent-adsorbate system obeying the Langmuir adsorption isotherm, $b = 0.48 \text{ bar}^{-1}$ and $p = 0.16 \text{ bar}^{-1}$. At what pressure will 50% of the surface be covered?